DLA: Unexpected phenomena in Mendelian disorders

GNET DLA_Before Mendelian Inheritance: X-linked inheritance

Two parts:

A. X-linked disorders
B. Unexpected phenomena in Mendelian disorders

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Required pre-reading: DLA: X-linked inheritance

- DLA 12a: X-linkage examples
- DLA 12b: Unexpected findings in Mendelian inheritance (Shown with examples)

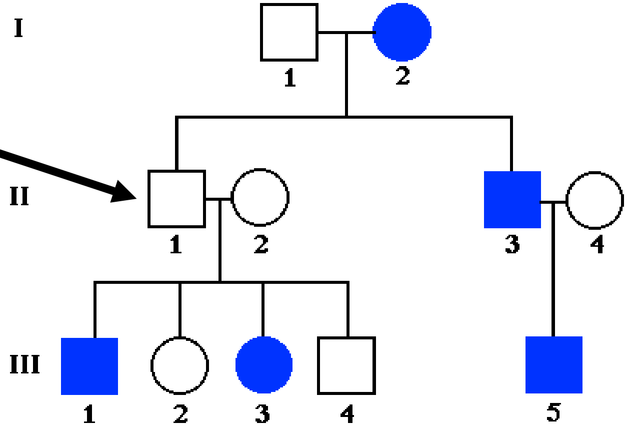
SOM.MK.III.BPM1.1.FTM.3.GNET.0037	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following X-linked disorders: Hemophilia A and B, G6PD deficiency, Duchenne and Becker muscular dystrophy, red/green color blindness. Lesch Nyhan syndrome, X-SCID as examples of X-linked disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0038	Summarize and distinguish the characteristics of the following patterns of inheritance using pedigrees: autosomal dominant, autosomal recessive, X-linked dominant and X-linked recessive
SOM.MK.III.BPM1.1.FTM.3.GNET.0039	Demonstrate how disease liability is transmitted for X-linked diseases
SOM.MK.III.BPM1.1.FTM.3.GNET.0040	Propose how the basic risk calculations are performed for X-linked disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0041	Explain how a woman who is a carrier of an X-linked disorder might manifest symptoms of the condition (manifesting heterozygote) Skewed X-inactivation, non-random X–inactivation
SOM.MK.III.BPM1.1.FTM.3.GNET.0042	Discuss X-linked dominant disorders and traits
SOM.MK.III.BPM1.1.FTM.3.GNET.0043	Compare and contrast variable expressivity and incomplete penetrance; Give examples of each
SOM.MK.III.BPM1.1.FTM.3.GNET.0044	Describe pleiotropy
SOM.MK.III.BPM1.1.FTM.3.GNET.0045	Compare and contrast locus heterogeneity and allelic heterogeneity; give examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0046	Describe locus heterogeneity in terms of complementation groups
SOM.MK.III.BPM1.1.FTM.3.GNET.0047	Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders

Objectives: Flipped class: X-linked inheritance

SOM.MK.III.BPM1.1.FTM.3.GNET.0048	Demonstrate how disease liability is transmitted for autosomal dominant diseases, autosomal recessive diseases and X-linked diseases
SOM.MK.III.BPM1.1.FTM.3.GNET.0049	Distinguish the different characteristics of recessive and dominant X-linked traits using pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0050	Identify obligate carriers in a pedigree depicting an X-linked trait
SOM.MK.III.BPM1.1.FTM.3.GNET.0051	Compare and contrast between X-linked, Y-linked, AR, AD, and mitochondrial traits.
SOM.MK.III.BPM1.1.FTM.3.GNET.0052	Discuss the concept of penetrance and variable expression and Differentiate between locus and allelic heterogeneity
SOM.MK.III.BPM1.1.FTM.3.GNET.0053	Calculate the recurrence risk based on penetrance for autosomal dominant and autosomal recessive disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0054	Define pleiotropy with examples. Be able to identify concepts if given a clinical scenario
SOM.MK.III.BPM1.1.FTM.3.GNET.0055	Recognize occurrence of new mutations as a cause of new genetic diseases in families. Be able to identify this concept from clinical scenarios and pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0056	Explain the phenomenon of germinal (germ-line) mosaicism
SOM.MK.III.BPM1.1.FTM.3.GNET.0057	Identify genetic disorders that have a delayed age of onset, and be able to identify this concept from clinical scenarios and pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0058	Explain the phenomenon of a manifesting heterozygote in X-linked recessive disorders, and be able to identify this concept from clinical scenarios and pedigrees

An Autosomal Dominant pedigree complicated by phenomenon of incomplete or reduced penetrance

Incomplete Penetrance



Genotype positive – Phenotype negative

- Male to male transmission (II-3 to III-5)
- But III-1 and III-3 do not have an affected parent
- II-1 **must have inherited the mutant allele** from I-2 and has passed on the mutant allele to III-1 and III-3
- II-1 most likely has the disease genotype but does not display the disease phenotype: Example of **Non-penetrance**
- A disorder is **fully penetrant**, if all the people carrying the mutation, express the phenotypic manifestations of the disorder

Incomplete penetrance

- Penetrance may be dependent on age, especially in adult-onset diseases, in which the proportion of people manifesting the disorder increases over time
- Called "***age dependent penetrance***"
  - Delayed age of onset
- Huntington disease has a high penetrance but a delayed age of onset!!
- Familial breast cancer has about 95% penetrance by the age of 80 years

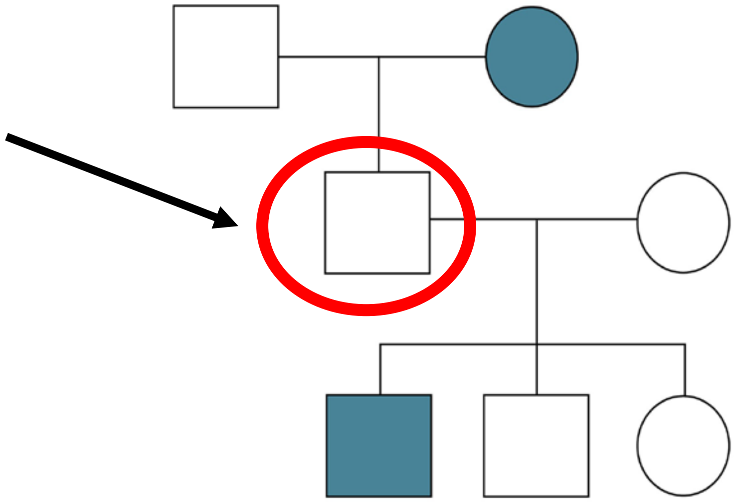


Figure 3.20 Example of nonpenetrance, in which an autosomal dominant trait is expressed in a grandparent and child but not in the parent. The unaffected parent must carry the mutant gene and is said to be nonpenetrant.

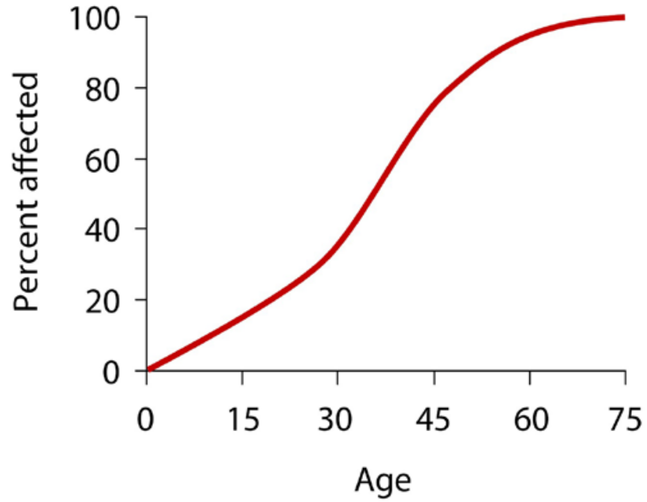


Figure 3.21 Age-dependent penetrance. The likelihood of being affected increases with age.

Calculation of recurrence risk based on penetrance

- Calculate risk of affected children for a couple (one person affected, other is normal)
- The penetrance of the autosomal dominant disorder is **80%**
- Remember, for autosomal dominant disorders: **50%** (½) recurrence risk for every child born to a couple if disorder is fully penetrant
- Risk = 40% risk of affected offspring when penetrance is 80%

$$\boxed{\frac{50 \times \mathbf{80}}{100} = 40\%\ \text{risk}}$$

- Calculation has to be modified for autosomal recessive disorders and X-linked disorders
- Autosomal recessive disorders may also show incomplete penetrance – Eg: Hemochromatosis

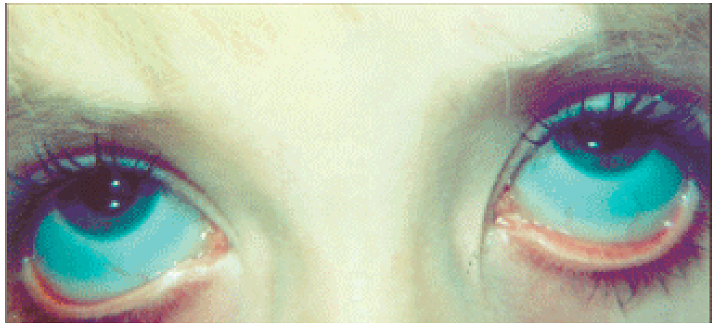
Variable Expression

- Some individuals are severely affected, and others are mildly affected
  - Variable phenotype
  - Sometimes observed in families (same mutant allele)
- Various reasons:
  1. Random chance
  2. Other genetic factors (modifier loci) or sex influence
  3. Type of mutation: Marfan syndrome
  4. Environmental exposure ….
- Hemochromatosis: Iron overload disorder (autosomal recessive disorder)
  - Hemochromatosis is more severe in males
  - Premenopausal females will menstruate and lose iron (blood)

More examples as we continue with this course

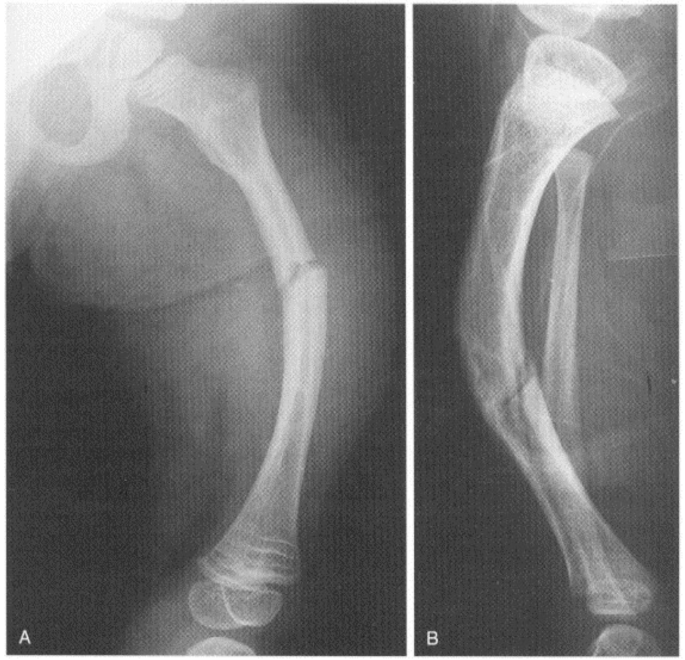
Variable Expression

A 2-year-old boy presents with deafness and multiple fractures. He was diagnosed with osteogenesis imperfecta (autosomal dominant). His older sister, a 12-year-old, girl has **blue sclerae**. She had one broken bone at the age of 7. Her father was similarly affected. **The younger sibling is more severely affected than the older sibling or the father (even though all of them have the same mutation)**



Blue sclerae in the older sibling (less severely affected)

Multiple fractures in the younger sibling (more severely affected)



Bone fractures in a patient afflicted with OI

Variable Expression

- Xeroderma pigmentosum (autosomal recessive):
  - More severe when exposed more frequently to **environmental** UV radiation (sunlight)
  - First child would be expected to be more severely affected than the second child
    - Parents might know to keep the second child out of the sun

Figure 2.12 Two children affected with xeroderma pigmentosum. Some children, such as the boy at the right, are more fair-skinned and sun-sensitive than others and develop freckling at a younger age. (Courtesy of Xeroderma Pigmentosum Society.)

Penetrance and variable expression

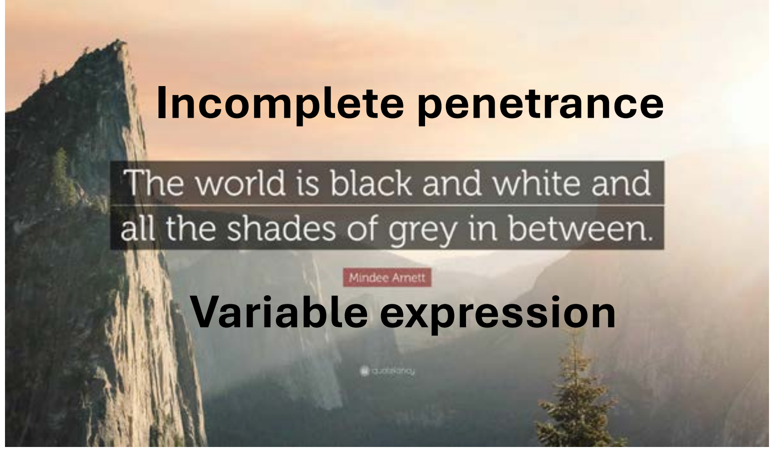
- Neurofibromatosis type I is an autosomal dominant disorder
- **High penetrance but variable expression**, even in the members of the same family
- Some have café-au lait spots, neurofibromas on the skin, axillary freckling, bone deformities, learning disabilities
- Generally, all patients with the mutant *NF1* gene show some signs of the disorder (**High penetrance**)

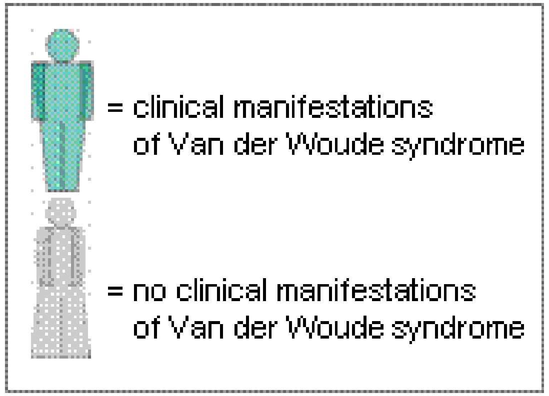
Incomplete penetrance and variable expression

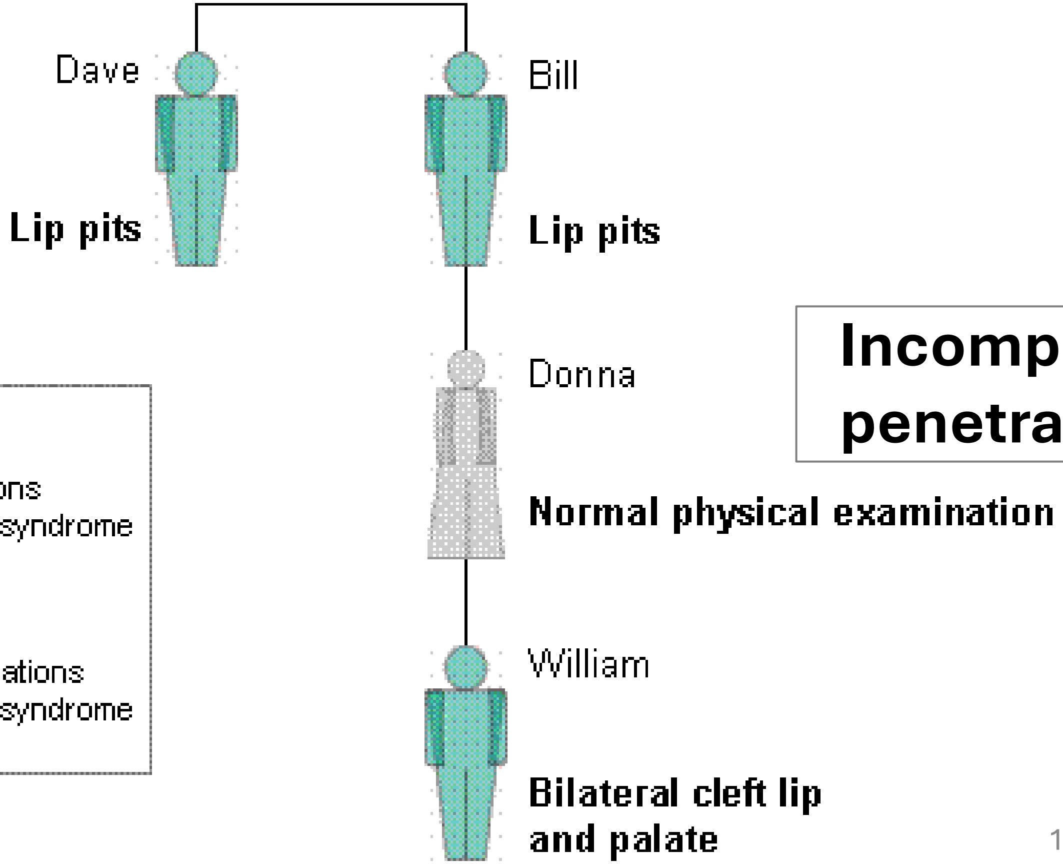
- Donna has the genotype for the autosomal dominant disorder, but NO phenotype of the disorder **(genotype positive-phenotype negative)**
- Calculate recurrence risk of the disorder in Donna's next child (given 60% penetrance)
- **Differentiate variable expression and incomplete penetrance**

$$\boxed{\frac{50 \times 60}{100} = 30\%\ \text{risk}}$$

Note: We don't expect you to memorize phenotype of Van der Woude syndrome or that it is AD







Incomplete penetrance

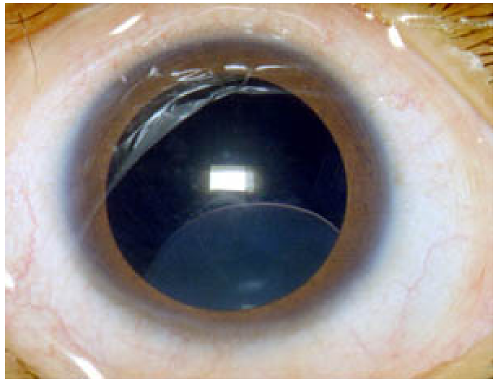
Pleiotropy

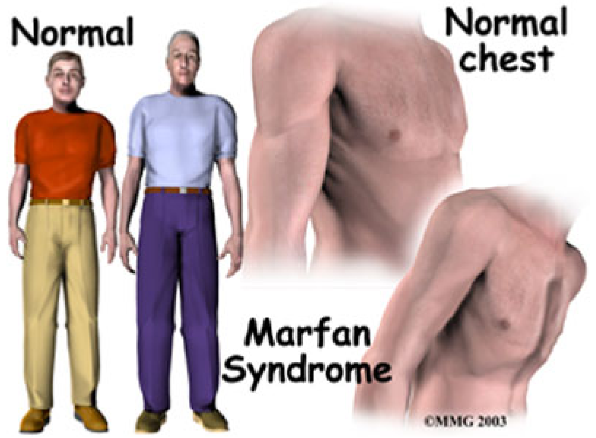
- A disease-causing mutation (single gene) affects more than one organ system
- **Marfan syndrome (autosomal dominant)**
  - Mutation in the **fibrillin-1 gene**
  - Skeletal abnormalities (arachnodactyly, long limbs, pectus excavatum)
  - Hypermobile joints
  - Ocular abnormalities (myopia, lens dislocation)
  - Cardiovascular disease (mitral valve prolapse, aortic aneurysm)
- **Osteogenesis imperfecta** (brittle bones, blue sclera) is due to a mutation in a gene encoding collagen.
- **Cystic fibrosis**: Respiratory infections and malabsorption

Marfan syndrome

- A 51-year-old man presents to the vascular surgery clinic for repair of an **iliac artery aneurysm.**
- The patient is a well-developed, well-nourished, **tall**, and **thin** man.
- Chest examination reveals a **tall, narrow thorax with outward protrusion of the chest wall**.
- Musculoskeletal examination reveals **long, thin fingers (arachnodactyly) and toes**.
- Eye examination shows **lens dislocation.**
- Chest radiography is performed as part of the routine preoperative evaluation.

Subluxation of lens





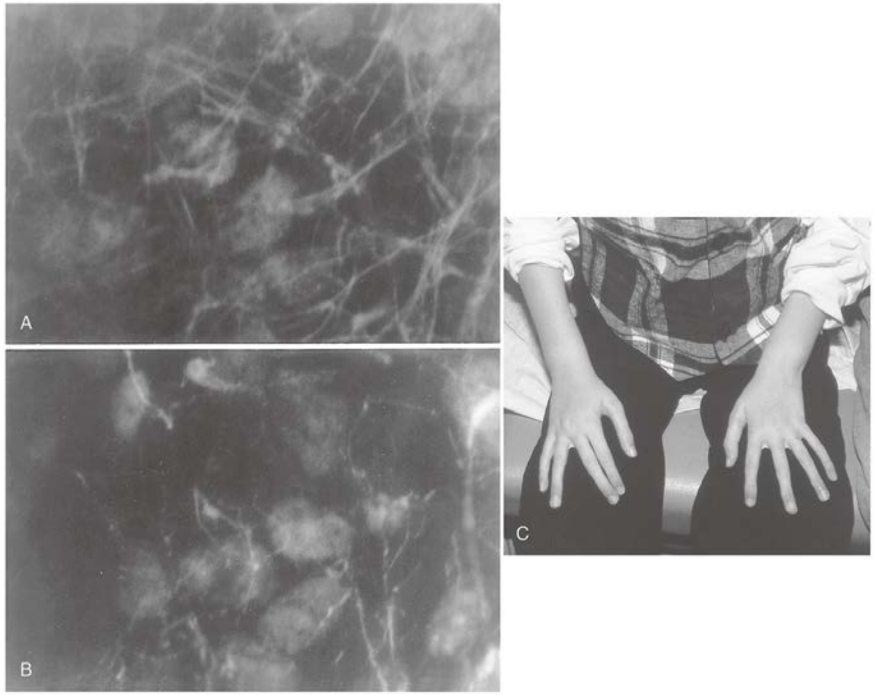


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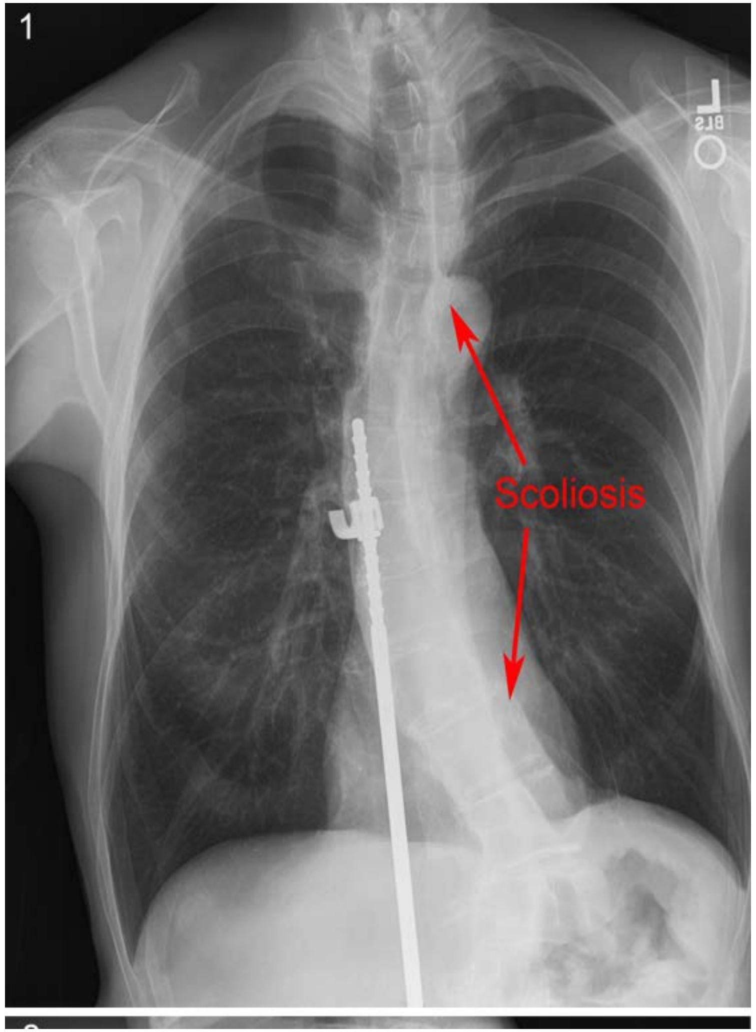
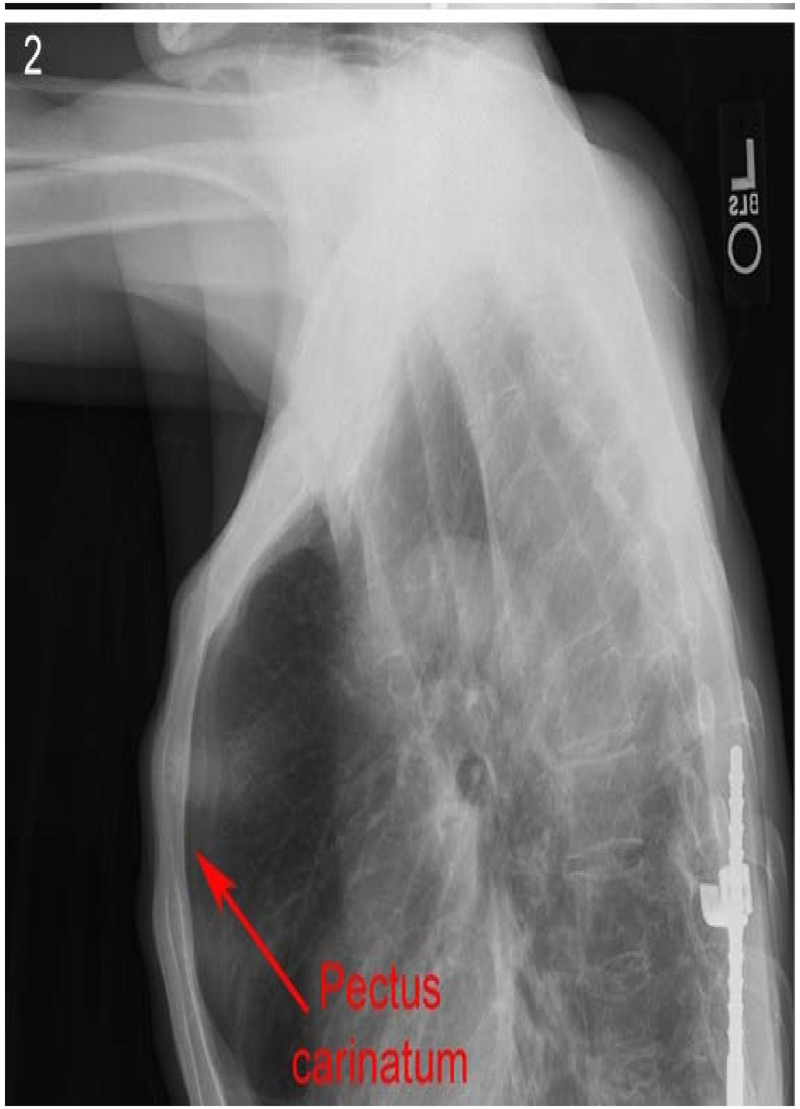
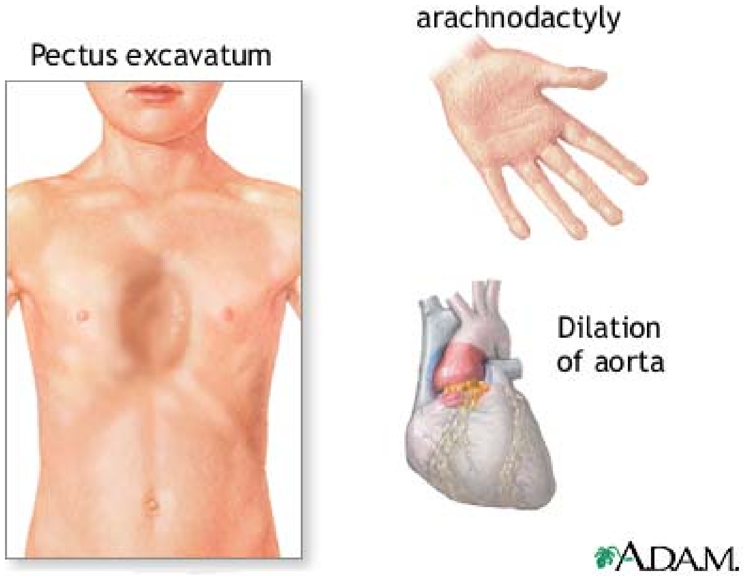
Figure 3.11 Immunofluorescence photomicrograph of normal (A) and Marfan syndrome (B) fibroblast sample stained for fibrillin, showing deficient fibrillin – 1 in the Marfan sample. (Courtesy of Dr. Heinz Furthmayr, Department of Pathology, Stanford University.) (C) Arachnodactyly seen in a child with Marfan syndrome. (Courtesy of Dr. Ronald Lacro, Department of Cardiology, Children's Hospital, Boston.)

Marfan syndrome: Clinical findings





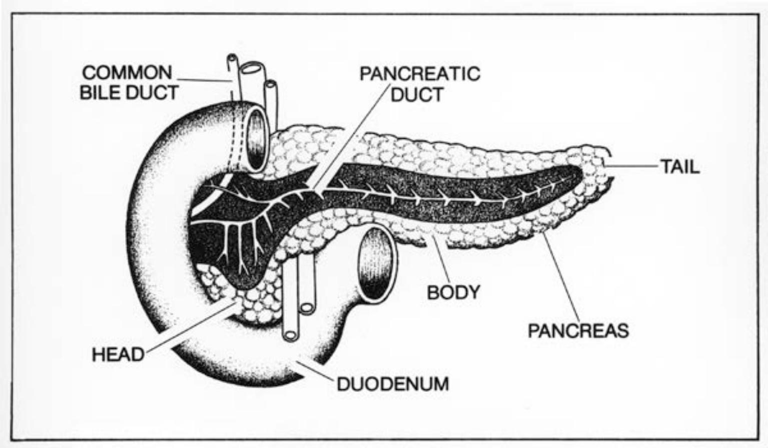


Cystic Fibrosis: Pleiotropy

- CFTR gene codes for a transmembrane gated-chloride channel
- Required for chloride secretion in exocrine glands → Maintains thin, watery mucus
- CFTR pathogenic variants → **Less chloride secretion** in pancreas and lungs → Viscous mucus → Blocks pancreatic ducts; Respiratory infections
- In sweat glands, CFTR **reabsorbs** chloride from sweat
- Laboratory test: Sweat chloride levels elevated

Pancreatic duct clogged by viscous mucus → Pancreatic insufficiency, and Malabsorption

Viscous mucus in the lungs not efficiently cleared from airways → Recurrent infection, inflammation, and lung tissue damage



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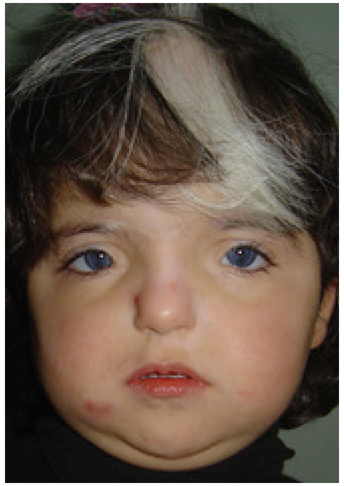
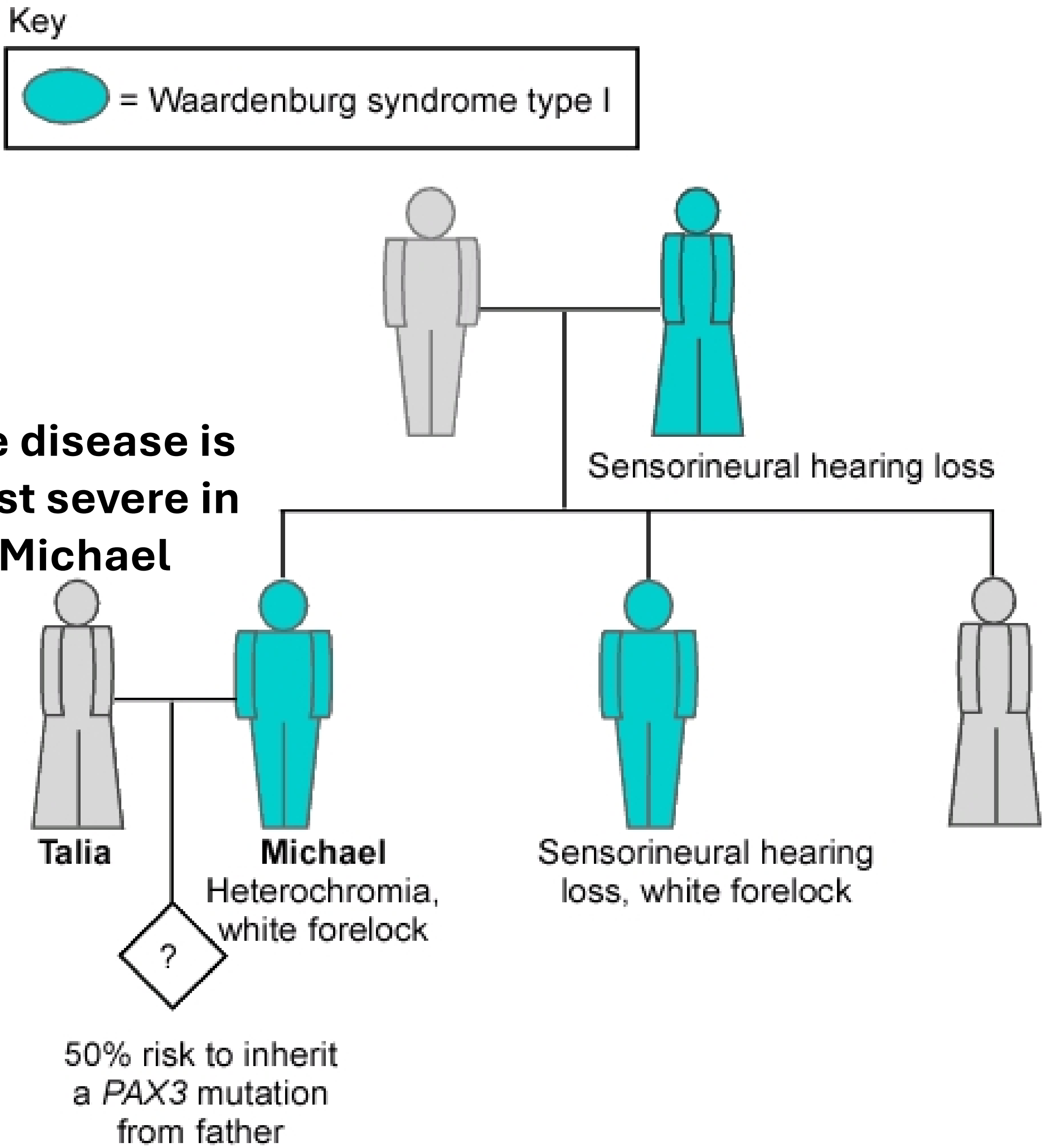
Pleiotropy and variable expression in Waardenburg syndrome

We don't expect you to memorize phenotypes or gene names for this syndrome

The disease is least severe in Michael





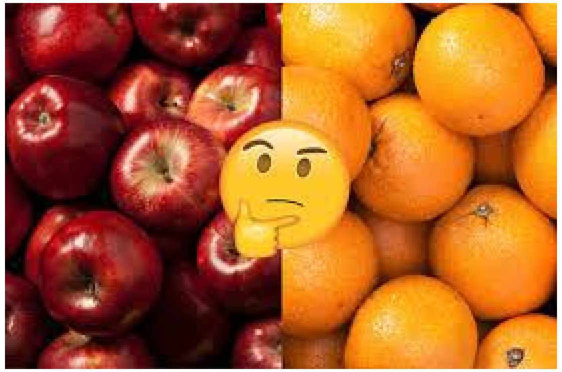
Locus heterogeneity

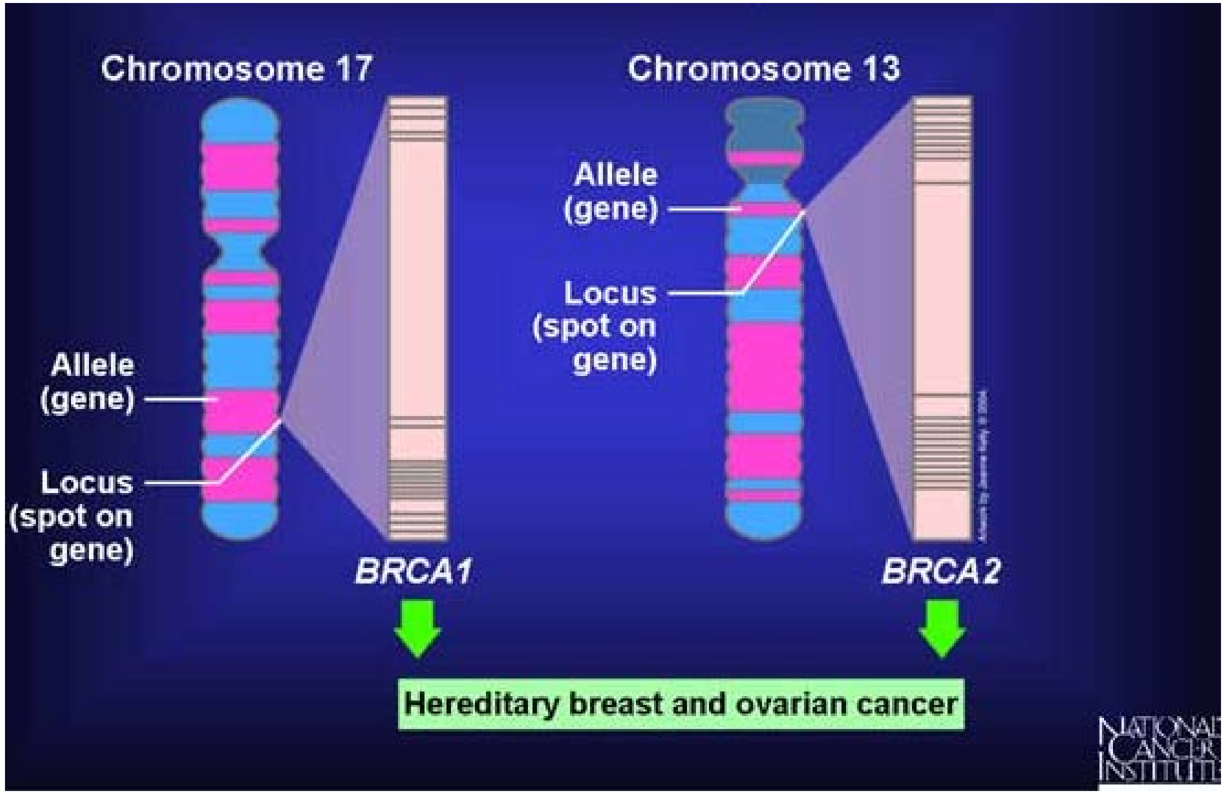
- Mutations at **different loci** that cause the same disease phenotype. Mutations of **different genes** cause the **same disease phenotype**.

- Osteogenesis imperfecta: Defect in collagen
  - Mutations of chromosome 17 (***COL1A1* gene**) or chromosome 7 (***COL1A2* gene**) lead to osteogenesis imperfecta

- Many disorders demonstrate locus heterogeneity (some examples):
  - SCID (AR and X-linked)
  - Familial hypercholesterolemia (LDLR or APOB)
  - Sensorineural hearing impairment
  - Charcot Marie Tooth disease (AD, AR, or X-linked)

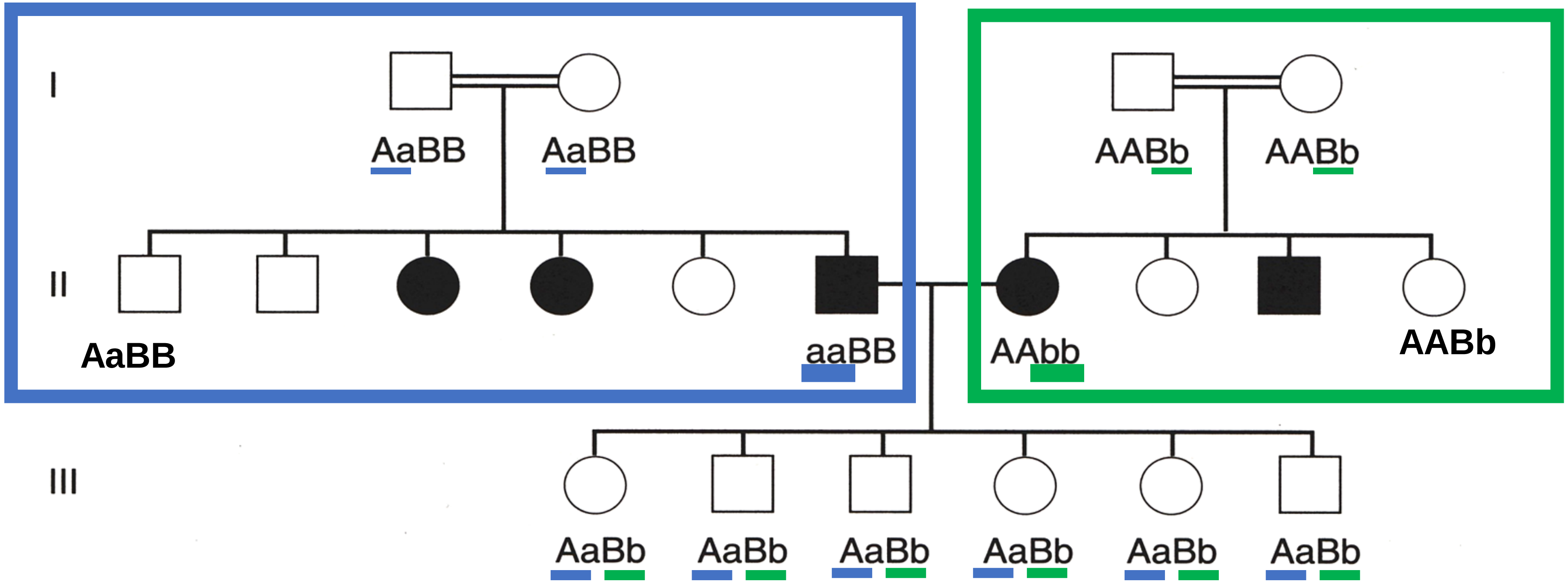
Breast cancer susceptibility genes- Locus heterogeneity





Mutations in *BRCA1* or *BRCA2* (two different genes at different locations) cause increased predisposition to hereditary breast cancer

Autosomal Recessive Inheritance & Locus Heterogeneity: Congenital Deafness



These children are not deaf because they are heterozygous at both loci.

A and B are two different genetic loci – each produce proteins required for normal hearing
a and b are the mutant (recessive) alleles of these genes

All children in Generation III are heterozygous for the two alleles for the two recessive loci.

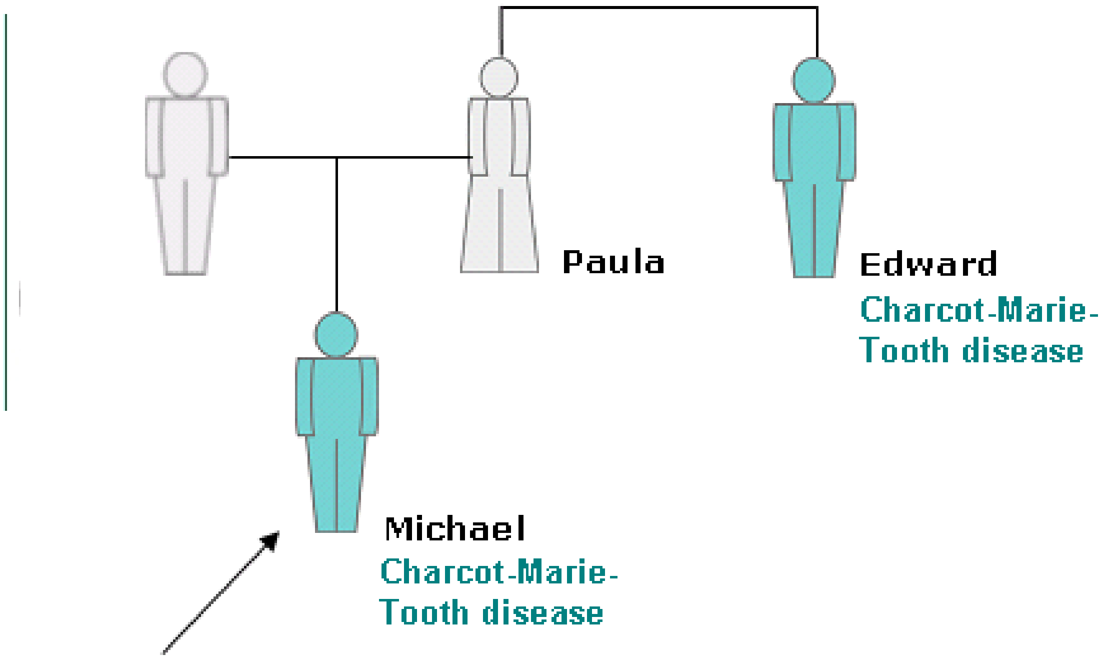
Over 15 different genes have been shown to be involved in autosomal recessive deafness (**Locus Heterogeneity):** Many genes can cause the same disorder

Autosomal Recessive Inheritance & Locus Heterogeneity: Congenital Deafness

- In generation-I all parents are carriers
  - I-1 and I-2 are carriers of the mutant ‘a’ gene
  - I-3 and I-4 are carriers of the mutant ‘b’ gene
- In generation-II, all individuals who have congenital deafness are homozygous for ‘a’ (II-3, II-4, II-6) or gene ‘b’ (II-7, II-9)
- In generation-II, all normal children are heterozygous for gene ‘a’ (II-1, II-2, II-5) or heterozygous for gene ‘b’ (II-8, II-10)
- In generation-III, none of the children are affected **(due to gene complementation)** as they are all carriers (heterozygotes) for gene ‘a’ and gene ‘b’
- Homozygosity for **gene ‘a’** or **gene ‘b’** can cause congenital deafness – **Locus heterogeneity**

Case: Charcot-Marie-Tooth (CMT) disease (Locus heterogeneity).

- Michael, a 16-year-old boy, is referred by a neurologist for genetic counseling regarding his recent diagnosis of CMT, a hereditary neuropathy. Michael's maternal uncle, Edward, is similarly affected, but Michael's mother, Paula, is asymptomatic.
- The geneticist explains to the family that several forms of CMT exist, all with similar clinical features. The most common form, CMT1A, is inherited in an autosomal dominant manner and caused by a duplication of a specific gene on chromosome 17; however, CMT1A is clinically indistinguishable from CMTX, an X-linked form. Molecular genetic testing later confirms that Michael has CMTX, which is consistent with his family history. CMT disease also has an AR inheritance.

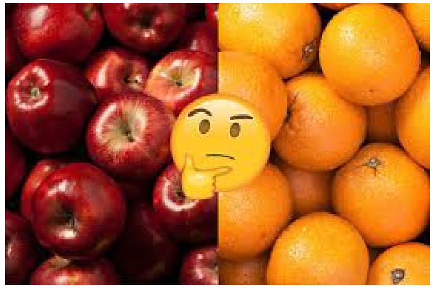


Reading – Clinical snapshot 2.2 on page 33 in Korf and Irons

Allelic heterogeneity

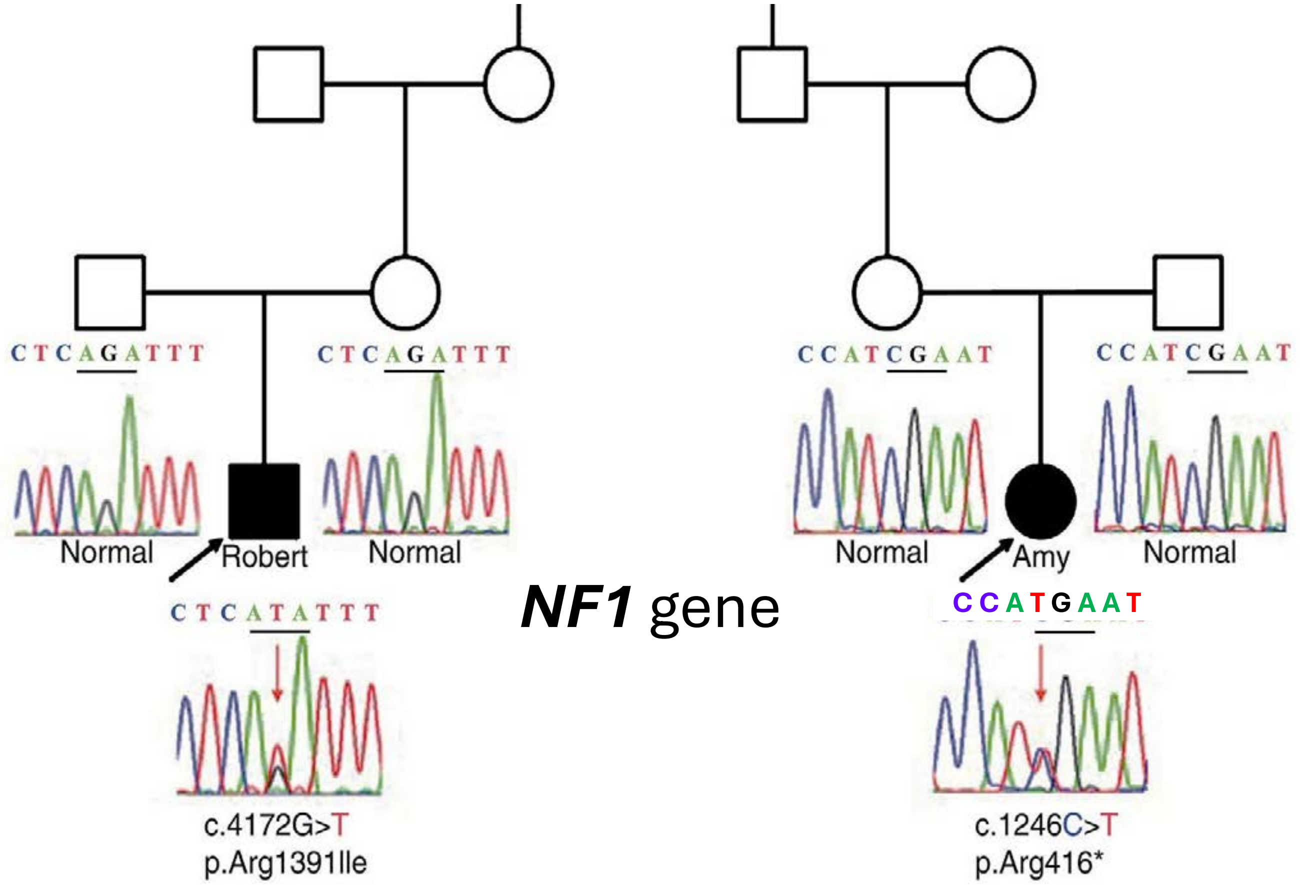
- Different kinds of mutations of the **same gene** cause the disease:
  - Neurofibromatosis-1 is caused due to a mutation in the ***NF1*** gene
    - 1000s of different mutations have been identified in the ***NF1*** gene.
  - Cystic fibrosis: **CFTR** mutations
    - Most common mutation is deletion of codon for Phe 508 (ΔF508) of *CFTR gene*
    - However, hundreds of other *CFTR* gene mutations reported (missense, frameshift, nonsense, promoter, etc)

- **Distinguish allelic heterogeneity vs locus heterogeneity**



Allelic heterogeneity of the ***NF1*** gene

- DNA sequencing of *NF1* gene shows different mutations of the *NF1* gene in Robert and Amy (note the differences in sequence)
  - Substitution mutation (Robert) in NF1
  - Nonsense mutation (Amy) in NF1
- NF1 gene has many mutation hot spots
- Many patients with neurofibromatosis may have a less common, or familial mutation (exome sequencing or gene sequencing).
- Compare this to sickle cell disease which is due to a Glu6→Val substitution in the beta-globin protein (no allelic heterogeneity)



NF1 gene

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Figure 10.5 Sequencing results for Robert (left) and Amy (right), showing two *de novo* and distinct *NF1* mutations.

Allelic heterogeneity resulting in a compound heterozygote in cystic fibrosis

= Carrier of an abnormal allele at a given locus

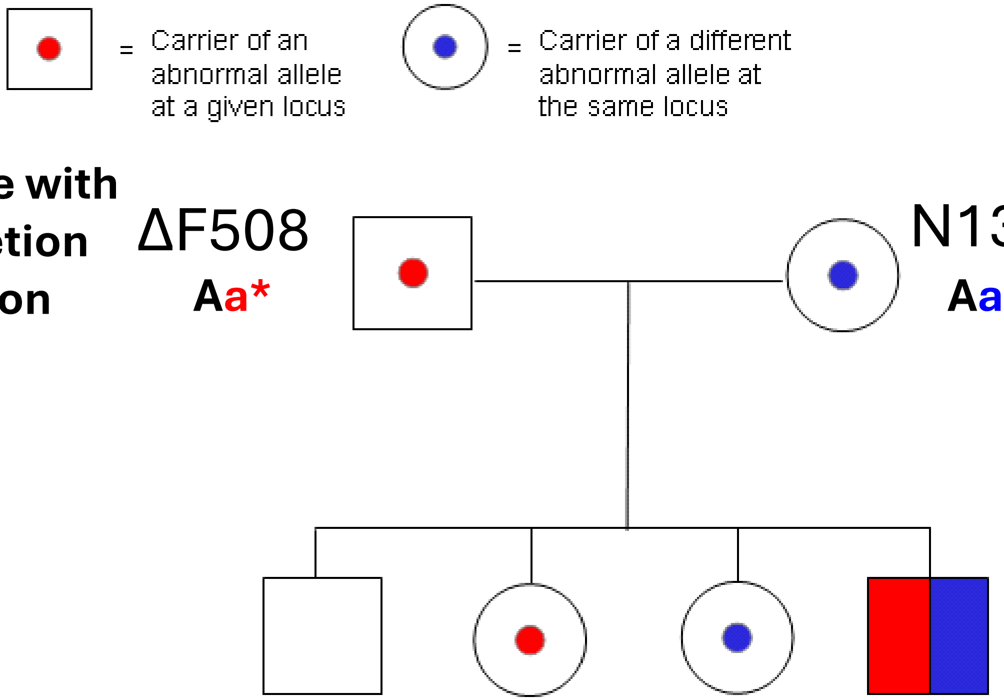
= Carrier of a different abnormal allele at the same locus

CFTR gene with Phe deletion mutation

ΔF508
Aa*

N1303K
Aa^

CFTR gene with an amino acid substitution mutation

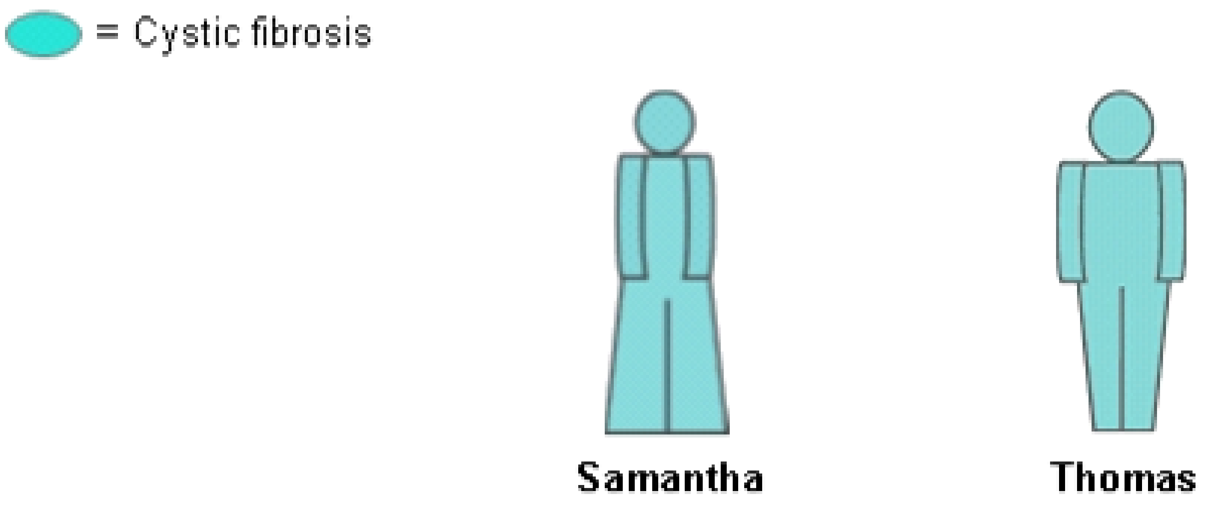


a*a^

Status	Unaffected, non-carrier	Carrier	Carrier	Affected, **compound heterozygote**
Mechanism	Inheritance of neither mutation	Inheritance of only the paternal mutation	Inheritance of only the maternal mutation	Inheritance of both the paternal and maternal mutations

Affected child has one CFTR allele with ΔF508 (deletion) and the other CFTR allele with N1303K (substitution)
Both CFTR alleles are pathogenic (non-functional) – but different types of mutation in CFTR gene

Compound heterozygote in cystic fibrosis



Lab test: **Elevated sweat chloride levels**

Phenotype:

- Meconium ileus at birth
- Chronic obstructive lung disease with frequent infections
- Pancreatic insufficiency leading to poor weight gain

Genotype (*CFTR* mutations)

- ΔF508 / N1303K
- G542X / R553X

Both mutant non-functional CFTR alleles

- Samantha and Thomas have **elevated sweat chloride levels** and phenotypic manifestations of cystic fibrosis (respiratory infections and malabsorption)
- Genetic testing shows allelic heterogeneity of CFTR gene **(compound heterozygotes)**

Summary:

- Recognize genetic principles in Mendelian disorders
- Differentiate unexpected phenomena in Mendelian disorders
  - Incomplete penetrance
  - Variable expression

  - Locus heterogeneity
  - Allelic heterogeneity
    - Compound heterozygote

  - Pleiotropy

Please email shupadhya@sgu.edu if you have any questions